PHASE — anti_bluff Submodule: Independent Review (§ / § / §)

Field	Value
Revision	1
Last modified	2026-07-22T23:36:54Z
Reviewer	(T1/main - claude1) independent review seam — NOT the build subagent
Review sub-strate (§11.4.209)	**Fable** (claude-fable-5), xhigh effort — Fable available, no fallback used
Scope	`constitution/submodules/anti_bluff/` @ `1e193e10bf-be6653db9ad18bfadff0767710f413` + constitution incorporation commit `439b6a07ca2660ed438dfdbbc6f0661f51107fdd` + public repos `github.com/vasic-digital/anti_bluff` / `gitlab.com/vasic-digital/anti_bluff`
Method	Every load-bearing claim independently re-executed (§11.4.6 — the build report was NOT trusted); every reported absence control-needled (§11.4.201(7)(b)); mutation probes run on temp COPIES only (working tree untouched, verified clean before and after)

Verdict

NO-GO (this round) — **0 BLOCKING / 2 IMPORTANT / 4 NIT**. Per §11.4.134 the review re-arms after forward-only remediation; both IMPORTANT findings have small, mechanical, forward-only fixes (§11.4.113 — no history rewrite needed or proposed). The core engineering is sound and honestly evidenced: **no seam passed its own golden-bad**, the RED baseline is genuine and independently reproduced, the out-of-box test is real, the repos are secret-clean, and the incorporation is clean and ff-only on all six mirrors.

. What was independently verified (all PASS — captured in this session)

1. **Full suite re-run, twice (§11.4.50):** `test/run_all.sh` → 41 PASS / 0 FAIL / exit 0, both iterations; verdict lines byte-identical across my two runs AND byte-identical to the committed `test/evidence/GREEN_iter1.txt` / `GREEN_iter2.txt` (diff empty). Claimed per-suite counts verified by enumeration: needle 9 (A1,B1,C1,D1,E1,F1,G1,H1,H2), custody 18 (A1–A3,B1,C1,D1,E1,E2,F1,G1,H1–H5,I1,I2,J1), evidence-class 6 (A1–F1), out-of-box 8 (OB1–OB8) = 41.
2. **Every golden-bad genuinely refuses (observed live in MY runs, not the report's):** SOL-03 — injected see-nothing grep → `INSTRUMENT-BLIND(2)` never `CERTIFIED-ABSENT`; alternation query + bare-literal needle → `NEEDLE-CLASS-MISMATCH(3)`; 2 MB SIGPIPE payload read PRESENT via read-to-EOF; mid-line carrier does NOT satisfy an anchored query (negative control). SOL-01 — raw `sqlite3` un-evidenced terminal `UPDATE` → `CUSTODY-REFUSED` (trigger abort); identical RED/GREEN fingerprints → refused; un-staged `Reopened` → refused; `item_history` `UPDATE` / `DELETE` → refused; open in-progress item NOT flagged by the sweep (§11.4.201(1) false-positive guard); empty items table → `SWEEP-BLIND-OR-EMPTY(2)` never PASS. SOL-04 — grep transcript wearing a runtime label → `WRONG-LAYER`; fieldless label → `SHAPE-INCOMPLETE`; artifact evidence on a user-visible defect → `CLASS-INSUFFICIENT`; source-on-source ACCEPTED (negative control); unreadable evidence → `BLIND(2)`. **No seam passed its own golden-bad.**

3. **Tests are RED-capable, proven by 3 independent mutations on temp copies** (repo untouched; §11.4.115(F) observation-before-trust): (M1) stripping the `custody_terminal_refuse` trigger from the template → custody suite FAILs (B1 "seam does not bind" + G1 + H-family); (M2) downgrading needle.sh blind-detection `return 2` → `return 0` → needle suite FAILs (C1); (M3) removing the WRONG-LAYER grep-echo branch → evidence-class suite FAILs (B1). All three exits non-zero with precise finding lines.
4. **§11.4.224 RED-first (claim 2):** `test/evidence/RED_initial.txt` shows 4/4 suites RED exit 1 with per-suite missing-artifact reasons. I **independently reproduced the transcript byte-for-byte** by copying ONLY the test files into a clean temp dir (no implementations, no commits) and running `run_all.sh` — the diff against the committed transcript is empty. Combined with (3), the tests are RED-capable and not assertion-free; the §11.4.224(A) observation-before-trust bar (Rev-60 disambiguation: authorship order is not the condition) is met.
5. **Out-of-box hermetic test is REAL (claim 3):** re-run by me from the incorporated clone `constitution/submodules/anti_bluff/` → 8/8 PASS. It `git clone`s the COMMITTED tree into a throwaway consumer, runs the documented `install.sh --db consumer.db --init`, and proves via plain `sqlite3` (no wrapper, no mock) that a raw un-evidenced terminal `UPDATE` on the consumer's own DB is `CUSTODY-REFUSED`, that a full-custody write lands with a DB-written audit row, and that sweep/evidence-class/needle all run from the clone. An uncommitted file cannot pass it (clone of committed state only) — verified by construction in `test_out_of_box.sh:40`.
6. **Constitution incorporation (claim 4):** depth-1 — NO `.gitmodules` inside anti_bluff (zero nested own-org submodules); `helix-deps.yaml` present, §11.4.31-conformant (`schema_version`, `deps: []`, `transitive_handling.recursive: true`, `conflict_resolution: operator-required`); gitlink at constitution HEAD = `1e193e1` (== both mirror tips). **ff-only:** `439b6a07` has exactly ONE parent — `6fd244ebe51f` (the pre-add HEAD) — and stages exactly 3 paths by name (`.gitmodules`, `submodules/anti_bluff`, the PHASE4 record); no history rewrite anywhere. My own `git ls-remote` on ALL SIX constitution mirrors (gitflic, github/HelixDevelopment, gitlab/helixdevelopment1, gitverse, github/vasic-digital, gitlab/vasic-digital) returns `main == 439b6a07ca2660ed438dfdbbc6f0661f51107fdd`.

7. **Project-literal scan (claim 4), control-needled:** the operative literal set — `atmosphere`, `rk3588`, `/mnt/track`, serials `998fd36615e99484` / `66ff9c4f51f00ee7`, `orange.?pi`, `presenter`, `kinopoisk`, `mistiq`, `vader`, `SPK-` — is **clean** across the tracked tree (instrument proven seeing: needle `anti_bluff` → 3 files; ERE path proven: 4 hits on the custody file). One fixture-string exception found and graded NIT-1 below.
8. **Public-repo hygiene (claim 5):** secrets scan over the FULL history (2 commits, 20 unique paths enumerated from `rev-list --all`) across 10 credential pattern classes (`sk-`, `ghp_`, `glpat-`, `AKIA`, `xox[baprs]-`, private-key blocks, `CLAUDE_CODE_OAUTH`, `ANTHROPIC_API_KEY`, `password=`, `Authorization: Bearer`) — **clean**, with the instrument's sight proven by a known-present needle through the same path. `.gitignore` covers the §11.4.30 classes incl. `.env` / `.env.*` with `.env.example` carve-out. **LICENSE:** MIT (vasic-digital, 2026) present + declared in README §7 — appropriate for a public reusable engine. Visibility: GitHub `PUBLIC` (gh API JSON); GitLab public PROVEN by **unauthenticated HTTPS** `ls-remote` succeeding (a private repo would refuse) — matching the continuum/containers/docs_chain convention.
9. **`ls-remote` fingerprints (claim 6), my own run:** GitHub `refs/heads/main` = `1e193e10bfbe6653db9ad18bfadff0767710f413`; GitLab `refs/heads/main` = `1e193e10bfbe6653db9ad18bfadff0767710f413`; local HEAD = same. All three identical — the report's fingerprint table is accurate.
10. **Parse-clean (§11.4.67):** `bash -n` OK on all 11 scripts. Submodule worktree clean (`git status --porcelain` empty) before and after review — the reviewed files ARE the shipped files.

. Findings (ranked; every fix FORWARD-ONLY per § . .)

IMPORTANT- — INSERT-path bypass of the status-write seam; the "UNWRITABLE for every writer" claim is overbroad and the README honest-boundaries section omits it

- **Where:** `seams/status_custody/custody_triggers.sql:22-23` (T1 is `BEFORE UPDATE OF status` only), `:41-43` (T2 same), `:54-56` (T3 same); `install.sh:12-14` ("a terminal status without a registered guard + RED/GREEN verdict pair ... is UNWRITABLE — for every writer, including raw sqlite3"); `custody_triggers.sql:2-6` ("bind EVERY writer"); `README.md:94-109` (§4 honest boundaries enumerates DROP TRIGGER, fabricated fields, oracle calibration, Obsolete, concurrency — but NOT this path); `PHASE4_ANTIBLUFF_APPLIED.md:29` (golden-bad table implies universal raw-writer binding).
- **Captured evidence (my live probe, this session):** on a fully-triggered DB (5/5 triggers verified), `INSERT INTO items(atm_id,title,type,status) VALUES('REV-001','insert bypass probe','Bug','Fixed');` was **ACCEPTED** — un-evidenced terminal status written, **zero** `item_history` rows. The sweep then caught it (`CUSTODY-FINDING[C1]` + `CUSTODY-FINDING[C4]` , exit 1) — defense-in-depth holds at Seam B, but the preventive write seam is silent.
- **Failure scenario (§11.4.194(1) — un-enumerated input dimension):** any import/migration/sync/backfill writer that INSERTS rows already in terminal status — exactly the raw-DB-write class §11.4.146(D3) exists for — lands an un-evidenced "Fixed" with no refusal and no audit row. Detection is deferred to the next sweep run, and **consumer sweep wiring is OWED** (README §5.5), so an out-of-box consumer that only ran `install.sh` has NO active layer on the INSERT path. The write-seam scenario space has two entry dimensions (UPDATE, INSERT); only one was covered and the other was assumed away — the exact §11.4.194 forbidden gap ("verifying one and assuming the rest").
- **Forward fix:** (a) add `BEFORE INSERT` twins of T1/T2 (+ either an `AFTER INSERT` audit row for status≠default or a documented exclusion) in `custody_triggers.sql` ; bump `apply_custody.sh` /sweep C0 trigger census

5→7; (b) add paired test cases — golden-bad (INSERT-with-terminal refused) + negative control (INSERT `'In progress'` accepted); (c) until (a) lands, amend `install.sh:12-14`, `custody_triggers.sql:2-6`, and README §4 to name the INSERT path as a known boundary. New commits on both mirrors, ff-only, then constitution gitlink bump.

IMPORTANT- — § . . coverage floor neither measured nor honestly tracked as owed

- **Where:** `README.md:111-126` (§5 OWED register) + `PHASE4_ANTIBLUFF_APPLIED.md:112-140` (§6 OWED) — neither carries the §11.4.224(B)/(E) code-coverage measurement for this NEW executable bash corpus (11 scripts). No percentage was claimed anywhere (honest — no fabricated figure, §11.4.6), but §11.4.224(E) is explicit: where the measurement is not run, the floor must land as an §11.4.3/§11.4.69 SKIP-with-reason + a tracked §11.4.197 item — "NEVER a silently-dropped floor". Here it is silently absent from an otherwise thorough OWED register that does track §11.4.85 stress/chaos and CM-gate wiring.
- **Corroborating measurement (mine, control-needled — instrument proven seeing via a 2-line needle child; suite ran GREEN under instrumentation; 5,306 trace lines):** `BASH_ENV`-propagated `PS4` line-trace during the full suite yields executed-unique-line lower bounds of 39%–66% per file (needle.sh 60%, ab_sql_lib 61%, apply_custody 45%, custody_sweep 48%, evidence_class_check 66%, install.sh 39%). **Honest instrument limits (§11.4.6):** LINE not branch coverage; multi-line commands/heredocs trace only their first line so the numerator systematically undercounts against a raw non-comment-line denominator — these figures are evidence that a REAL calibrated measurement is owed, NOT a proven sub-85% verdict. (First measurement attempt returned an all-zero result; per §11.4.201(7)(b) the zero was treated as INSTRUMENT-BLIND, re-instrumented, and needle-proven before any figure above was recorded.)
- **Forward fix:** add "§11.4.224 coverage measurement (kcov or calibrated PS4-trace) + per-corpus threshold calibration" to README §5 + the PHASE4 record §6 as a tracked §11.4.197 item; run it in the next slice.

NIT- — ATMOSphere-derived fixture string vs the "ZERO project literals" claim

- **Where:** `test/test_evidence_class.sh:49` — `RUNTIME_OBSERVABLE: grep: 5 hits for RouteToSecondary in VideoOutputManagerService.java` . `VideoOutputManagerService` + `RouteToSecondary` are ATMOSphere-internal framework names inside a golden-bad fixture of a PUBLIC repo whose README header and `helix-deps.yaml:21` claim "ZERO project literals". Functionally inert (§11.4.201(7)(a) — a mention, not a binding; the engine's behavior is fully project-agnostic, confirmed by the clean operative-literal scan), but it contradicts the claim's letter and leaks a private project's internal class name into a public repo. (`ATM-001` -style fixture ids are NOT flagged — `ATM-NNN` is the §11.4.54 universal anchor's own canonical example.)
- **Forward fix:** genericise the fixture string (e.g. `RouteFeature in SomeVideoService.java`) in a follow-up commit.

NIT- — `nq_stream_contains` certifies absence without a control needle

- **Where:** `lib/needle.sh:86-100` . The stream variant closes the SIGPIPE false-absent class (proven) but reports `ABSENT` with no needle argument — the blind-instrument class that file-mode `nq_absent` refuses (`return 2`) is uncovered in stream mode (a blind `NQ_GREP` yields count 0 → `ABSENT(1)` , not `BLIND(2)` ; only producer failure is detected). README §4's "measurements not routed through needle.sh stay exposed" covers un-routed measurements, not this in-library asymmetry.
- **Forward fix:** accept an optional needle after the query (mirroring `nq_absent` semantics) OR name the boundary explicitly in the file header + README §4.

NIT- — PHASE record header timestamp is a rounded placeholder

- **Where:** `PHASE4_ANTIBLUFF_APPLIED.md:6` — `Last modified: 2026-07-23T00:00:00Z` ; the incorporation commit's real UTC time is `2026-07-22T23:21:19Z` . §11.4.44 wants the real edit timestamp (same Minor class as the Rev-63 M1 remediation precedent). Forward fix on next touch.

NIT- — incorporated clone's origin pushes to GitHub only

- **Where:** `constitution/submodules/anti_bluff` local config — `origin` push URL = GitHub only, while the sibling `continuum` submodule's origin fans to GitHub + GitLab; a future push from THIS clone would reach one mirror (§2.1 multi-upstream norm). No `upstreams/` recipe dir ships, so §11.4.36 auto-wiring cannot apply. (Both mirrors DID receive the current HEAD — this concerns future pushes from the incorporated clone only.)
- **Forward fix:** add the GitLab push URL to origin in the incorporated clone (local config, no commit) and/or ship an `upstreams/` recipe dir next slice.

. Summary for the conductor

Question	Answer
Did any seam pass its own golden-bad?	**NO** — all golden-bads refuse, verified live in my own runs + 3 mutation probes on temp copies all caught
RED-first genuine?	**YES** — committed RED transcript independently reproduced byte-for-byte from tests-without-implementations
Out-of-box test real?	**YES** — 8/8 from the incorporated clone; real `git clone` of committed state, real raw-`sqlite3` refusal on the consumer DB
Incorporation clean?	**YES** — depth-1, valid `helix-deps.yaml`, operative-literal set clean (control-needled), ff-only single-parent commit on `6f-d244e`, all 6 mirrors at `439b6a07`
Public repos secret-clean + licensed?	**YES** — full-history 10-class scan clean (needled); MIT LICENSE; PUBLIC verified on both hosts (GitLab via unauthenticated HTTPS)
Fingerprints	**CONFIRMED** — `1e193e10bfbe6653db9ad18bfad-ff0767710f413` on local + GitHub + GitLab (my own `ls-remote`)
Verdict[4]	**NO-GO this round** — 0 BLOCKING / 2 IMPORTANT / 4 NIT; forward-only remediation, then re-review to GO per §11.4.134

. Anti-bluff certification of this review

- Every PASS above was observed in MY OWN executions this session, never copied from the build report (§11.4.6); the suite was run twice with identical verdict lines (§11.4.50) and cross-diffed against the committed GREEN evidence.

- Every reported absence (literals, secrets) ran a control needle through the SAME instrument + path before being reported (§11.4.201(7)(b)); one of my own instruments went blind mid-review (first coverage attempt), was treated as INSTRUMENT-BLIND per the discipline, and was re-proven seeing before any figure was recorded.
- Mutation probes ran exclusively on temp copies; the submodule worktree was verified clean (`git status --porcelain` empty) — review-only constraints honored: no edits, no commits, no pushes, no repo changes; the forbidden files (`pre_build_verification.sh` , `meta_test_false_positive_proof.sh` , `docs/workable_items.db` , /mnt/track2-4) were not touched.
- Review substrate: Fable, xhigh effort (§11.4.209) — recorded above.